Memory and Forecasting, Version 2

Oliver Smirnov & Nikhil Kumar

Abstract

Participants watch a numeric AR(1) process one value at a time, with no graph, and then forecast its next value from memory alone. Sixty native English speakers aged 18–30 take part, half with persistence $\rho = 0.6$ and half with $\rho = 0.8$. Each sees 20 values and then makes 200 forecasts. This document describes the experiment as it is implemented: the task, the stimuli, the scoring and payment, and the data recorded.

1 Design at a glance

Participants	60 native English speakers, aged 18–30 (Prolific; desktop or laptop only)
Between-subjects factor	Persistence $\rho \in \{0.6, 0.8\}$, 30 participants each
Process	$x_t = 500 + \rho\,(x_{t-1} - 500) + \varepsilon_t$
Standard deviation of the process	20 in every condition (each series has SD exactly 20 in both phases)
Values per participant	220: 20 observation rounds, then 200 forecasting rounds
Range of values	integers from 400 to 600
Distractor tasks between rounds	none
Duration	about 30 minutes (as stated to participants)
Pay	\$4.00 flat fee plus a performance bonus (\$6.00 expected)

Table 1: Summary of the design.

Each value of the process is the previous value's deviation from 500, scaled by ρ, plus a random shock ε_t. The shock's standard deviation is set to $\sigma_\varepsilon = 20\sqrt{1-\rho^2}$, so that the process has the same standard deviation, 20, in every condition and persistence is the only thing that differs. Table 2 lists the resulting shock (innovation) standard deviation for each persistence level; more persistent processes have smaller shocks.

ρ	Innovation SD σ_ε
0.6	16.00
0.8	12.00

Table 2: Conditions (30 participants each).

2 Stimuli

One pre-generated series per participant. Seventy-two series of 220 integers are generated in advance and stored in a stimuli file, 36 per ρ. Sixty are needed (30 per condition) and the other twelve are spares, so that a participant who leaves can be replaced without anyone's series being reused. Each participant is assigned one series when they enter the study, so no series is ever shown to two participants. A new participant is placed in whichever condition currently has fewer finished or active participants, so the two conditions stay balanced as people drop out and are replaced, and a participant's condition is unrelated to who they are.

Start value. The first value is a draw from the process's stationary distribution, a normal distribution with mean 500 and standard deviation 20, rounded to an integer. It is not restricted, and it is never adjusted afterwards.

Properties of every series. The series are constructed so that their realized statistics equal the true values. A random AR(1) draw, especially its first 20 values, can have a realized persistence well away from the true ρ, and participants learn the process from exactly what they see. Each series is therefore adjusted in integer steps until it satisfies Table 3, separately in the observation phase (values 1–20), the forecasting phase (values 21–220) and the series as a whole.

Statistic	Value in every series
Persistence (Observation Phase)	within 0.0005 of ρ
Persistence (Forecasting Phase)	within 0.0005 of ρ
Persistence (Whole Series)	within 0.0005 of ρ
Mean (Observation Phase)	exactly 500
Mean (Forecasting Phase)	exactly 500
Standard Deviation (each phase)	exactly 20
Values (all)	400 to 600

Table 3: Properties of every generated series. Persistence is the OLS slope of x_t on x_{t-1} with an intercept; in the forecasting phase it uses the value seen before each forecast to predict the next. The standard deviation is the population form (dividing by the number of values).

Because the realized persistence equals the nominal ρ, a participant's implied persistence can be compared with either. Selecting series on their own sample statistics means they are not plain random draws of the process, and statistics that are not fixed above, such as autocorrelation at longer lags, can differ slightly from an unadjusted AR(1) series.

3 Procedure

Everything below happens in the participant's web browser. The task is a sequence of jsPsych screens; each text entry is submitted with the Return/Enter key only, and there is no on-screen submit button.

3.1 Fullscreen and attention check

After a consent form, the browser enters fullscreen mode. Every exit from fullscreen is counted and the participant is shown a warning. The participant then reads a short statement from the principal investigator that asks them to find a quiet room, not to take notes, and not to disrupt data quality, and that says data may be discarded if these conditions are not met. Two short questions follow:

1. The statement asks the participant to click the word "Blue" if they agree to the conditions (the alternative is "Orange").
2. A passage on memory (a quotation from Jane Austen) contains an embedded instruction to answer a later question with "never". The participant answers two memory-history questions, how old their oldest memory is and how often they realize they have forgotten something. The second must be answered "never".

A participant who misses *both* the "Blue" check and the "never" check is screened out: they see a short message and are sent to Prolific with a screen-out completion code.

3.2 Instructions

The instructions are two pages after a short welcome screen. The first (Figure 1, left) explains that the participant will forecast the values of a mathematical process that generates values between 400 and 600, one at a time; that each value depends on earlier values and on random fluctuation, with the exact relationship unknown to them; that after each forecast they see how accurate it was on a bar; and that more accurate forecasts earn more points and a larger bonus. The second page (Figure 1, right) says that the first few tasks will not require forecasts (they simply type the number shown), that they must not take notes, and that they may reread the instructions with the *Previous* button before pressing *Begin Experiment*.

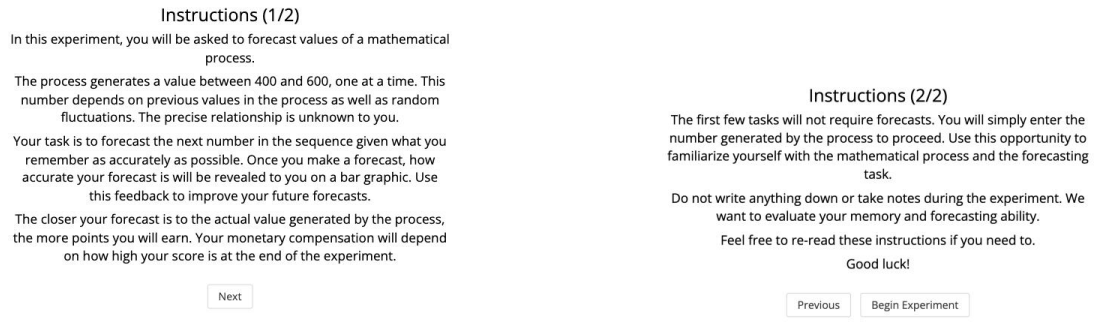


Figure 1: Instruction pages 1 and 2.

3.3 Observation phase (rounds 1–20)

For each of the first 20 values, the participant sees the current value in large type above a horizontal bar (Figure 2a). The bar spans the full range of the process, 400 on the left to 600 on the right, and is filled in blue up to the current value; tick marks run along it. The participant types the

value they see and presses Enter. If they type something else, a red message appears under the box ("That number is not valid. Please enter the number shown.") and they must try again. The correctly typed value is recorded, the next value appears, and no forecast is requested. The aim is to make sure each value is attended to and encoded, and the participant never sees a graph of the series, so everything they later forecast from has to come from memory. During this phase the last observation screen adds a brief note that the observation phase is ending and forecasting is about to begin.

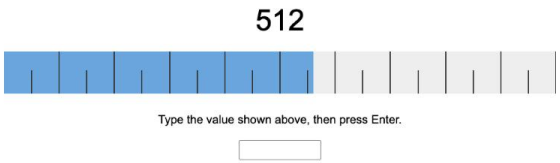
3.4 Forecasting phase (rounds 21–220)

Each of the remaining 200 rounds has two screens.

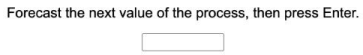
1. **Forecast.** The screen reads "Forecast the next value of the process, then press Enter" and shows an empty text box (Figure 2b). No previous values are shown. The forecast must be a three-digit whole number between 400 and 600. Anything else is refused with a red message under the box: "Enter a 3-digit number." for entries that are not three digits, and "Enter a number between 400 and 600." for three-digit entries outside the range (Figure 2c).

2. **Feedback.** The true value appears in large type above the bar, filled in blue to that value, with a red vertical line marking the participant's forecast (Figure 2d). The distance between the red line and the end of the blue fill is the forecast error. A box in the corner shows the running score. As in the observation phase, the participant types the true value shown and presses Enter. This keeps the new observation actively encoded before the next forecast.

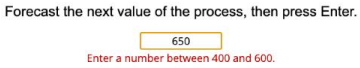
A forecast is scored once per round (Section 4), even if the feedback screen has to be retyped.



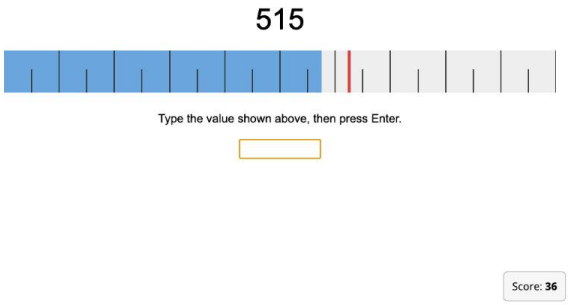
(a) Observation: type the value shown.



(b) Forecast screen.



(c) Out-of-range forecast: error message, no progress.



(d) Feedback: true value (blue fill), the participant's forecast (red line) and the running score.

Figure 2: The four screens that make up the task.

3.5 Results and comments

After the last round, the participant sees their final score, their flat payment and their bonus (Figure 3). They are then asked for optional free-text comments on anything confusing or any technical problems, leave fullscreen, and are redirected to Prolific with the completion code.

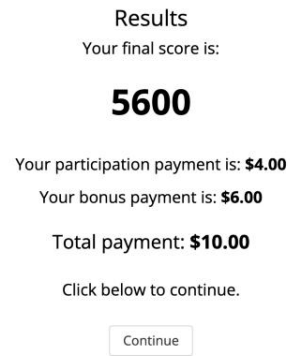


Figure 3: Results screen, shown for a score at the 5,600-point benchmark.

4 Scoring and payment

Points per forecast. Each forecast earns a whole number of points between 0 and 100, falling linearly with the absolute error up to a cutoff of 20 (the process's standard deviation):

$$\text{points} = \text{round}\left(100 \cdot \max\left(0,\ 1 - \frac{|\text{forecast} - \text{actual}|}{20}\right)\right).$$

A perfect forecast earns 100 points, an error of 10 earns 50, and an error of 20 or more earns 0. Over the 200 forecasting rounds the highest possible score is 20,000.

Bonus. The bonus is proportional to the total score, with no cap: \$6.00 for a score of 5,600, which is the pilot study's average of 28 points per forecast over 200 forecasts. The flat fee is \$4.00, so a participant who scores 5,600 earns \$10.00, about \$20 per hour at the stated 30 minutes. The largest possible bonus, for a perfect score of 20,000, is about \$21.43. The server recomputes the score and the bonus from the recorded trials; the number the browser displays is used only as a cross-check.

What participants can expect to earn. The table below gives average scores on the 30 series in each condition for two benchmark strategies: always using the best linear forecast, $500 + \rho(x_{t-1} - 500)$, which requires knowing the process, and always answering 500.

ρ	Best forecaster: score (SD)	Bonus	Always 500: score	Bonus
0.6	8,833 (309)	\$9.46	7,412	\$7.94
0.8	10,883 (255)	\$11.66	7,307	\$7.83

Table 4: Average scores over 200 forecasts on the experiment's series, and the bonus each would earn.

Participants are not told the process and must infer it, so their bonuses will typically be much lower than the best forecaster's. Because the scoring cutoff is the same in both conditions, a forecaster who knows the process earns more at $\rho = 0.8$ than at $\rho = 0.6$.

5 Data recorded

Trial data are sent to the server throughout the task, so a participant who drops out still leaves the trials they finished. The server stores, for every participant: the Prolific identifiers, the assigned series and ρ, the status (started, completed, screened out), the attention-check outcome, the number of fullscreen exits, the time of each trial, the number of attempts, the browser-reported score and the server-recomputed score, the bonus, and the free-text comments. For each trial it stores the screen type, round index, the true value, the typed value, whether the typed value was correct, and (for forecasts) the forecast, the error and the points earned.

6 Primary measure

The main quantity is each participant's *implied persistence*: the slope from regressing their forecast on the value they had just seen, over the 200 forecasting rounds. It is compared with the persistence of the series they saw, which equals ρ to within 0.0005, to give an excess-persistence, or overreaction, measure, and the measure is compared across the two conditions. Participants who were screened out are excluded; participants who leave without finishing are not analyzed.

7 Distractors (not included)

The software can run a timed word-judgment task between rounds, but this experiment does not use it. When switched on, each round is preceded by a short block of about four seconds in which participants answer as many yes/no questions as they can about single words (for example, "Will this item fit in a shoebox?" or "Does this item refer to something living?"), pressing the right Shift key for yes and the left Shift key for no (Figure 4). The task would add an instruction page and a mention in the consent text.



Figure 4: Word-judgment screen (not used in this experiment).